the same collateral (priority).[199] On the whole, this statute establishes a functional framework generally applicable to all secured transactions principles, while also incorporating bespoke rules that address the unique characteristics and practices associated with different types of collateral. It balances theoretical elegance with commercial pragmatism.[200]

The Article 9 taxonomy was originally developed in the 1950s and has been periodically amended to accommodate emerging asset classes and commercial practices. Unsurprisingly, prior to the 2022 UCC Amendments, this architecture did not include any classification specifically designed for digital assets. As a result, digital assets fell within the residual category of "general intangibles"—a catch-all designation that has historically served as the default for novel intangible assets, from intellectual property rights to franchise agreements.[201]

As a result, prior to the 2022 UCC Amendments, the regime for use as collateral of digital assets could be summarized as follows. For creation (attachment) of a security interest, three requirements had to be met: the secured party had to give value, the debtor needed rights in the collateral, and the parties had to enter into an authenticated security agreement describing the collateral.[202] Regarding perfection, secured parties had only one option: filing a financing statement in the appropriate public registry.[203] Unlike with certain other types of collateral, Article 9 did not provide alternative perfection methods such as possession or control.[204] As for priority, the general rule of "first-to-file-or-perfect" governed competing claims, meaning that the first secured party to either file a financing statement or otherwise perfect their security interest through another available method had priority over subsequent secured

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[199] U.C.C. § 9-317.

[200] *See* GRANT GILMORE, SECURITY INTERESTS IN PERSONAL PROPERTY (1965); Grant Gilmore, *The Secured Transactions Article of the Commercial Code*, 16 LAW & CONTEMP. PROBS. 27 (1951); Grant Gilmore, *The Good Faith Purchase Idea and the Uniform Commercial Code: Confessions of a Repentant Draftsman*, 15 GA. L. REV. 605 (1981); Grant Gilmore, *Article 9: What It Does Not Do for the Future*, 26 LA. L. REV. 300 (1966).

[201] "General intangible" means any personal property, including things in action, other than accounts, chattel paper, commercial tort claims, deposit accounts, documents, goods, instruments, investment property, letter-of-credit rights, letters of credit, money, and oil, gas, or other minerals before extraction. The term includes payment intangibles and software. *See* Schroeder, *supra* note 170 at 9-15, 19-21 (explaining why digital assets, including Bitcoin, would not fit in the categories of UCC Article 9 "money" and "deposit accounts").

[202] U.C.C. § 9-203(b)(1)-(3)(A).

[203] U.C.C. § 9-310(a).

[204] *See* U.C.C. § 9-312(b) (listing perfection methods for various types of collateral but providing no alternative to filing for general intangibles)

parties.[205] Moreover, security interests would automatically attach to digital assets received as proceeds of original collateral and remain perfected.

Thus, prior to the 2022 UCC Amendments, the framework for using digital assets as collateral was both clear and well understood. Market participants could rely on long-standing Article 9 principles for their secured transactions.[206] The system accommodated blanket liens on all present and future digital assets of a debtor, permitted straightforward perfection through filing, and provided clear priority rules based on temporal filing order. This predictability and simplicity, in principle, allowed parties to structure digital asset financing with relative confidence.

Regrettably, this regime was fundamentally misaligned with market realities, commercial practices and technology of digital assets. Perfection through filing—the sole method available—was particularly problematic. The slow pace and inherent time lag associated with public registration systems stood in stark contrast to the near-instantaneous execution of digital asset transfers and the frenetic pace of their trading environments. Furthermore, the pseudonymity of DLT networks severely impaired both the identification of appropriate filing jurisdictions and the searching for existing liens. Even more telling was the emergence of arrangement that ignored the Article 9 framework entirely. In these transactions, debtors would transfer their digital assets directly to lenders as security for loans, with the mutual understanding that the assets would be returned upon repayment. Typically consummated with minimal documentation and no public filings, these dealings reflected secured parties' preference for taking control over digital assets collateral, rather than relying on registries. By the early 2020s, both scholars and practitioners had come to view that the status quo was unsustainable.[207]

3. Tokenizations

As described in Part I, the market for digital assets extends beyond cryptocurrencies like Bitcoin and Ether that participants trade for their intrinsic value and technological utility. A significant segment of the DLT ecosystem centers on tokenization: using digital assets, typically NFTs, to purportedly

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[205] U.C.C. § 9-322(a)(1).

[206] Odinet, *supra* note 89.

[207] *See* Schroeder, *supra* note 170 at 67; Ronald J. Mann, *Reliable Perfection of Security Interests in Crypto-Currency*, 21 SMU SCI. & TECH. L. REV. 159, 172-73 (2018); Xuan-Thao Nguyen, *Lessons from Case Study of Secured Transactions with Bitcoin*, 21 SMU SCI. & TECH. L. REV. 181, 182-83 (2018); James P. Nehf, *Security Interests in Virtual Currencies*, (March 2, 2020), https://ssrn.com/abstract=3547540; TriBar Opinion Comm., *Third-Party "Closing" Opinions Under the 2022 Amendments to the Uniform Commercial Code*, 79 BUS. LAW. 555 (2024), https://www.americanbar.org/groups/business_law/resources/business-lawyer/2024-spring/tribar-report-on-opinions-under-2022-amendments/.

represent ownership of other assets—both tangible and intangible—or rights enforceable against persons, such as money claims.[208]

Tokenization emerged several years after Bitcoin's creation, made possible by the capabilities of the Ethereum network.[209] The first major breakthrough came with CryptoKitties in 2017, a project that enabled holders of NFTs to appropriate, control, and trade unique digital cats.[210] Building on the success of this experiment, enthusiasts started advocating for the use of digital assets to represent a variety of assets, not merely digital cats.[211]

The market's response was explosive. By August 2021, monthly NFT trading volume reached an unprecedented $2.8 billion.[212] The art world led the charge, exemplified by Christie's landmark $69 million sale of Beeple's digital artwork in March 2021.[213] Sports leagues joined the movement, with NBA Top Shot generating over $800 million in trading volume from tokenized basketball highlights.[214] Entrepreneurs rushed to tokenize an ever-expanding range of assets—from real estate and fine art masterpieces to luxury automobiles—with platforms like Propy, Maecenas, and CurioInvest claiming their tokens represented fractional ownership in these physical assets.[215]

Amid this whirlwind of excitement and grandiose predictions about transforming ownership, commercial transactions and society as a whole, a small group of legal scholars and practitioners questioned the private law foundations of tokenization.[216] They posited that, as a matter of private law, these digital assets did not embody ownership rights or personal claims enforceable against third parties—despite issuers' proclamations to the contrary.[217] These isolated voices denounced a profound disconnect between marketing promises, popular assumptions and the law.[218]

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[208] *See* Moringiello & Odinet, *supra* note 25
[209] *See id.*
[210] *Id.*
[211] *Id.*
[212] *Id.*
[213] *Id.*
[214] *Id.*
[215] *Id.*
[216] *Id. See also* Odinet & Tosato, *supra* note 25
[217] *See* Moringiello & Odinet, *supra* note 25; *see also* Odinet & Tosato, *supra* note 25
[218] *See* Moringiello & Odinet, *supra* note 25; *see also* Odinet & Tosato, *supra* note 25.

In our view, these critics were correct. Prior to the 2022 UCC Amendments, digital asset tokenizations were almost invariably flawed.[219] This becomes evident upon examining the applicable private law framework.[220]

For all the revolutionary rhetoric surrounding digital assets tokenizations, the concept of representing rights in one thing through another has been a cornerstone of commercial law for centuries.[221] Legal systems have long recognized certain paper documents as embodiments of rights: negotiable instruments carry payment rights, share certificates represent fractional corporate ownership, deeds embody property interests in land, bills of lading and warehouse receipts confer title to goods in transit and storage respectively.[222]

Through centuries of development, the law has crafted sophisticated frameworks for each of these tokenization mechanisms. These regimes establish clear rules ensuring both transacting parties and affected third parties understand their rights and obligations. Their complexity stems from their unique position at what Thomas Merrill and Henry Smith call the "property/contract interface."[223] Traditional tokenization mechanisms blend contractual and proprietary elements, each governed by distinct rules. The law affords parties extensive freedom in shaping their respective rights and obligations in contract, while property-based dimensions are subject to strict constraints.[224]

For the purpose of our inquiry, it is important to highlight that property law constraints tokenizations in two fundamental respects. First, it limits what types of personal property can be tokenized. For instance, payment rights are embodied in negotiable instruments, goods in transit are represented by bills of lading, and stored commodities are tokenized through warehouse receipts.[225] These categories are not left to private discretion but are instead determined by law.[226] For example, one cannot deposit corporate securities certificates in a warehouse and seek to obtain a warehouse receipt representing ownership of those securities, just as one cannot convert a barber's promise of a future haircut into a negotiable check.

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[219] *See* Moringiello & Odinet, *supra* note 25; *see also* Odinet & Tosato, *supra* note 25.

[220] *See* Moringiello & Odinet, *supra* note 25; *see also* Odinet & Tosato, *supra* note 25.

[221] *See* Moringiello & Odinet, *supra* note 25.

[222] *See id.*

[223] Thomas W. Merrill & Henry E. Smith, *The Property/Contract Interface*, 101 COLUM. L. REV. 773 (2001).

[224] *Id.*

[225] *See* Moringiello & Odinet, *supra* note 25.

[226] *See id.*